Q29

$$E = \frac{1}{2}mv^2 \Rightarrow p = \sqrt{2mE}$$

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}} \propto \frac{1}{\sqrt{m}}$$

Double slit formula: $\Delta y = \dfrac{\lambda D}{d}$

The fringe spacing Δy for double-slit experiment is proportional to λ, thus

$$\Delta y \propto \frac{1}{\sqrt{m}}$$

$$\frac{(\Delta y)_n}{(\Delta y)_{C_{60}}} = \sqrt{\frac{m_{C60}}{m_n}} = \sqrt{720}$$

Ans: B

Q30